

A strongly normalizing pure type system with ranks as sorts and Type-in-Type

Pavel Sokolov

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Abstract

We present a dependently typed pure type system which contains a circular axiom $s : s$ for a certain sort s and is nonetheless strongly normalizing. Girard's paradox is avoided because our sorts are not closed under a Π -former: function types are placed in sorts according to their *rank*. A normalization proof is surprisingly simple (modulo general metatheoretical properties of pure type systems) and has been fully formalized in vanilla Rocq without any additional axioms.

- 1 Pure type systems and Girard's paradox
- 2 PTS-rk and its strong normalization
- 3 Formalizing PTS-rk in Rocq